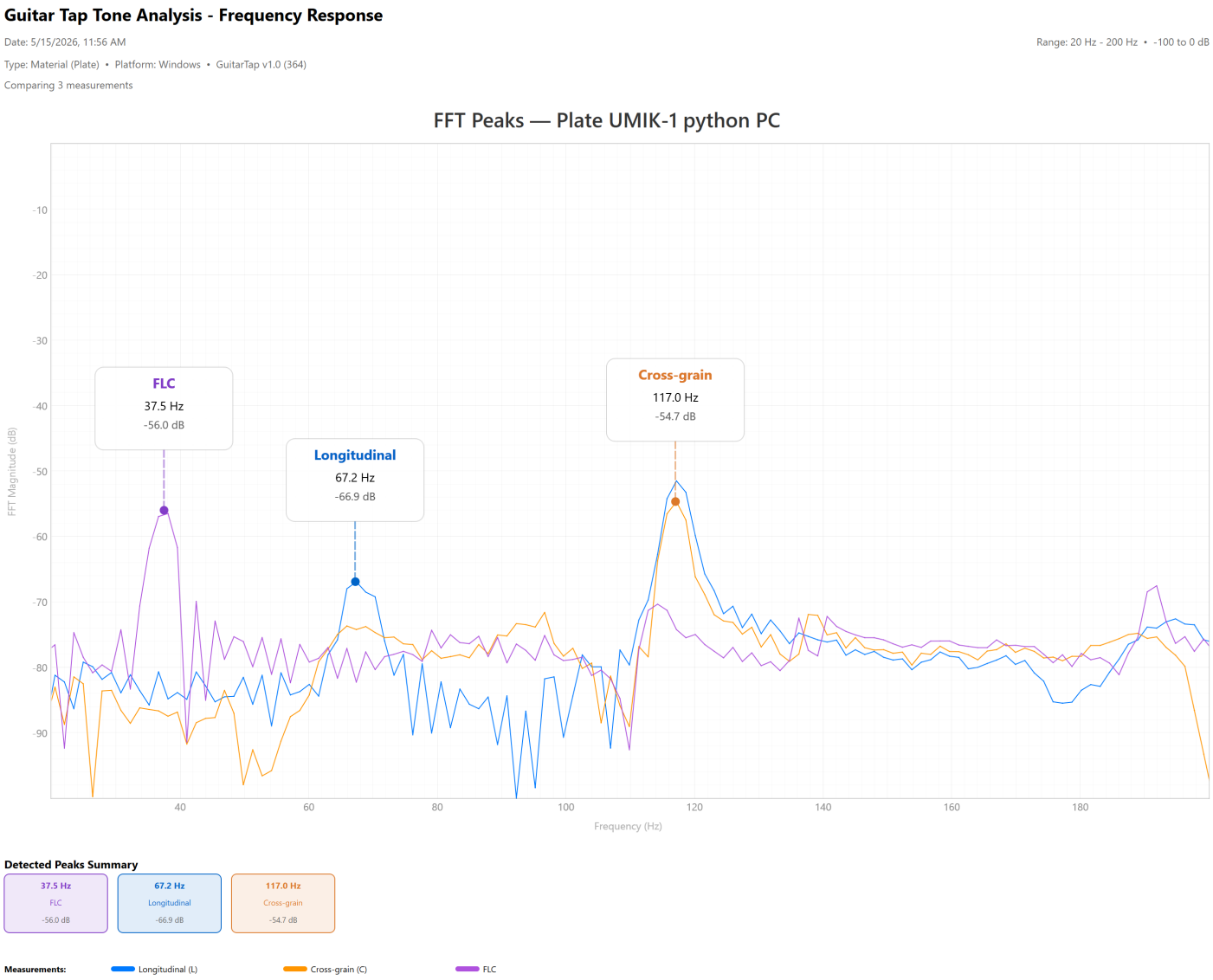
GuitarTap

Tap Tone Analysis Report

May 15, 2026
11:56 AM

Measurement Name:	Plate UMIK-1 python PC
Type:	Material (Plate)
Frequency Range:	20.0 Hz – 200.0 Hz
Microphone:	Microphone (Umik-1 Gain: 18dB) · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
37.5 Hz	-56.0 dB	D1	8.7	FLC (Diagonal)

67.2 Hz	-66.9 dB	C2	9.2	Longitudinal (L)
117.0 Hz	-54.7 dB	A#2	20.0	Cross-grain (C)

Gore Target Thickness

2.86 mm

Body: 490 × 390 mm · f_vs = 60 (Classical Top)

GLC (Shear Modulus): 0.383 GPa

Plate Properties

Sample Dimensions

Length: **557.5 mm** Width: **220.5 mm** Thickness: **4.85 mm**

Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **67.2 Hz** fC: **117.0 Hz** fLC: **37.5 Hz**

Speed of Sound (L): **4192 m/s**

Speed of Sound (C): **1142 m/s**

Young's Modulus (L): **6.13 GPa**

Young's Modulus (C): **0.45 GPa**

GLC (Shear Modulus): **0.383 GPa**

Specific Modulus (L): **17.6** (Fair)

Specific Modulus (C): **1.3** (Very Good)

Radiation Ratio (L): **12.0**

Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.074**
typical: 0.04 – 0.08

Long/Cross Ratio: **13.5**
typical: 12 – 25

Overall Quality: Good

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.